DSA0601 DATA HANDLING AND VISUALIZATION
CO1 AT2 Data Interpretation Assessment (R Programming)
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Question 1: Mapping Data onto Aesthetics

Dataset

Product	Sales (₹ Lakhs)	Profit (%)	Category
P1	120	18	Electronics
P2	90	12	Clothing
P3	150	25	Electronics
P4	80	8	Furniture
P5	135	22	Furniture
P6	100	16	Clothing

Questions

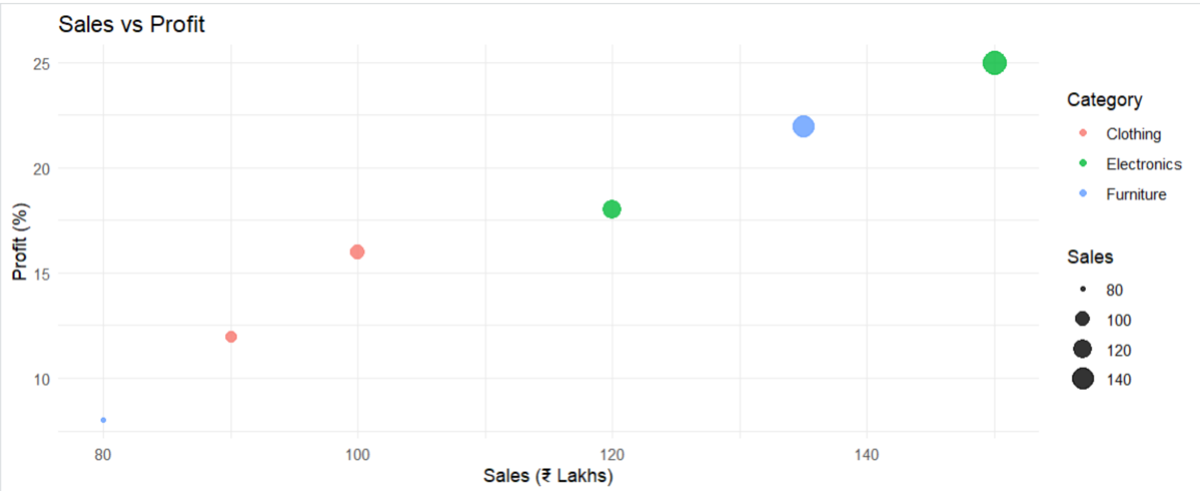
1. Write an R program to create a scatter plot with Sales on the x-axis and Profit on the y-axis.

R Code

```
library(ggplot2)

# Create dataset

data <- data.frame(

  Product = c("P1","P2","P3","P4","P5","P6"),

  Sales = c(120,90,150,80,135,100),

  Profit = c(18,12,25,8,22,16),

  Category = c("Electronics","Clothing","Electronics","Furniture","Furniture","Clothing")

)

# Scatter Plot

ggplot(data, aes(x = Sales,

            y = Profit,
```

```
            color = Category,
            size = Sales)) +
  geom_point(alpha = 0.8) +
  labs(title = "Sales vs Profit",
     x = "Sales (₹ Lakhs)",
     y = "Profit (%)") +
  theme_minimal()
```

Result



A scatter plot is generated with:

- **Sales** on the X-axis.
- **Profit** on the Y-axis.
- Different **colors** representing product categories.
- Point **sizes** representing the sales value.

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2. Map Category using an appropriate aesthetic and Sales using another suitable aesthetic.

Answer

In the scatter plot, different aesthetics are used to represent different variables effectively.

- **Category** is mapped to the **Color** aesthetic.
  - Electronics, Clothing, and Furniture are displayed using different colors.

    - This helps distinguish products belonging to different categories easily.
- **Sales** is mapped to the **Size** aesthetic.
    - Products with higher sales are displayed using larger points.
    - Products with lower sales appear as smaller points.
    - This allows the viewer to understand sales magnitude without reading the exact values.

Using both color and size simultaneously improves the readability of the visualization and enables comparison of multiple variables in a single chart.

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3. Interpret the relationship between Sales and Profit.

Answer

The scatter plot indicates a **positive relationship** between Sales and Profit.

- Products with higher sales generally have higher profit percentages.
- Product **P3** has the highest sales (150 Lakhs) and also the highest profit (25%).
- Product **P5** also shows relatively high sales (135 Lakhs) with a high profit of 22%.
- On the other hand, Product **P4** has the lowest sales (80 Lakhs) and also the lowest profit (8%).

Although there are slight differences among products, the overall trend suggests that increasing sales is associated with increased profit.

Therefore, the visualization indicates a **positive correlation** between Sales and Profit.

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4. Identify the type of scale used for each variable.

Answer

The dataset contains both continuous and categorical variables.

Variable	Data Type	Scale Used	Explanation
Sales	Numerical	Continuous Scale	Sales values are numeric and can take any value within a range.
Profit	Numerical	Continuous Scale	Profit percentages are continuous numerical values.

Variable	Data Type	Scale Used	Explanation
Category	Categorical	Discrete (Categorical) Scale	Category contains distinct groups such as Electronics, Clothing, and Furniture.
Product	Categorical	Discrete Scale	Product IDs are labels used for identification only.

Continuous scales are suitable for numerical variables, whereas categorical scales are used for variables containing distinct groups.

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5. Suggest one alternative visualization that better represents the same data.

Answer

One suitable alternative visualization is a **Bubble Chart**.

Reasons

- A Bubble Chart can represent four variables simultaneously.
- The **X-axis** can represent Sales.
- The **Y-axis** can represent Profit.
- Bubble **size** can represent Sales or another numerical variable.
- Bubble **color** can represent Category.

Another possible visualization is a **Grouped Bar Chart**, which compares sales and profit values across different product categories. However, a Bubble Chart provides a more effective representation because it shows relationships among multiple variables in a single visualization.

Therefore, the **Bubble Chart** is the most suitable alternative visualization for this dataset.

Question 2: Coordinate Systems (Cartesian and Polar)

Dataset

Direction	Frequency
N	18
NE	12
E	20
SE	15
S	22
SW	11
W	16
NW	14

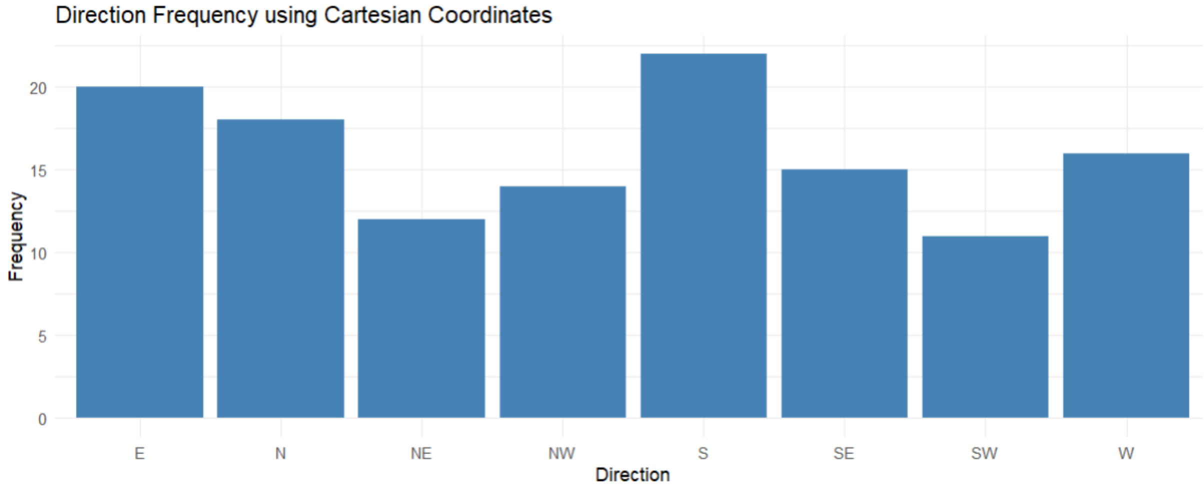
Questions

1. Write an R program to visualize the dataset using a Cartesian coordinate system.

R Code

```
library(ggplot2)

# Create dataset

data <- data.frame(

 Direction = c("N","NE","E","SE","S","SW","W","NW"),

 Frequency = c(18,12,20,15,22,11,16,14)

)

# Cartesian Bar Chart

ggplot(data, aes(x = Direction, y = Frequency)) +

 geom_bar(stat = "identity", fill = "steelblue") +

 labs(title = "Direction Frequency using Cartesian Coordinates",

    x = "Direction",

    y = "Frequency") +

 theme_minimal()
```

Result



A **Cartesian bar chart** is displayed with directions on the X-axis and frequencies on the Y-axis. Each bar represents the frequency of a particular direction.

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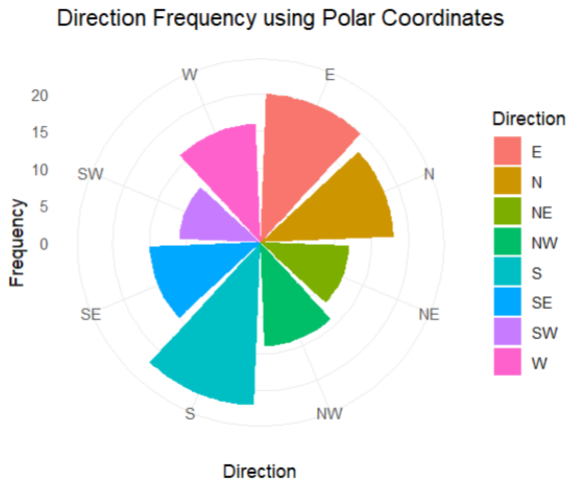
2. Write another R program to visualize the same data using a Polar coordinate system.

R Code

```
library(ggplot2)

# Create dataset

data <- data.frame(

  Direction = c("N","NE","E","SE","S","SW","W","NW"),

  Frequency = c(18,12,20,15,22,11,16,14)

)

# Polar Chart

ggplot(data, aes(x = Direction, y = Frequency, fill = Direction)) +

  geom_bar(stat = "identity") +

  coord_polar() +

  labs(title = "Direction Frequency using Polar Coordinates") +

  theme_minimal()
```

Result



A **Polar chart (circular bar chart)** is displayed. Each direction is arranged around a circle, and the length of each segment represents its frequency.

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3. Compare the effectiveness of both visualizations.

Answer

Both the Cartesian and Polar coordinate systems present the same data but are useful for different purposes.

Cartesian Coordinate System

- Represents data using horizontal (X-axis) and vertical (Y-axis) axes.
- Makes it easy to compare the exact frequencies of different directions.
- Differences in bar heights are clearly visible.
- Suitable for numerical comparison and statistical analysis.
- Easier to read precise values from the graph.

Polar Coordinate System

- Represents data in a circular format.
- Directions are displayed around a circle, similar to a compass.
- Provides a natural representation of directional information.
- Makes it easier to identify directional patterns.
- More visually attractive but less accurate for comparing exact values.

Comparison

Cartesian Coordinate System	Polar Coordinate System
Easier to compare values	Better for directional data
Uses X and Y axes	Uses circular layout
High accuracy	Better visual appeal
Suitable for analysis	Suitable for presentation

Thus, the Cartesian chart is more effective for comparing frequencies, whereas the Polar chart is more effective for representing directional data.

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4. Explain which coordinate system is more appropriate for directional data.

Answer

The **Polar coordinate system** is more appropriate for directional data because directions naturally follow a circular arrangement.

For example:

- North (N)
- North-East (NE)
- East (E)
- South-East (SE)
- South (S)
- South-West (SW)
- West (W)
- North-West (NW)

These directions form a complete 360° circle. A Polar chart represents this circular nature more effectively than a Cartesian chart.

The Polar coordinate system is commonly used in:

- Wind direction analysis
- Compass navigation
- Weather forecasting
- Radar charts

- Circular statistical data

Therefore, Polar coordinates provide a more meaningful representation of directional information.

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5. Justify your answer based on data interpretation.

Answer

The Polar coordinate system is justified because it improves the interpretation of directional data.

- It preserves the natural order of compass directions.
- Adjacent directions appear next to each other, making patterns easier to recognize.
- The circular arrangement reflects real-world directional relationships.
- Users can quickly identify the direction with the highest or lowest frequency.

From the dataset:

- **South (S)** has the highest frequency (**22**).
- **East (E)** has the second-highest frequency (**20**).
- **South-West (SW)** has the lowest frequency (**11**).

These differences are clearly visible in both charts, but the Polar chart presents them in a way that matches how directional information is interpreted in real life.

Hence, the Polar coordinate system is the most suitable choice for this dataset.

Question 3: Color Scales for Data Representation

Dataset

Student	Marks
A	45
B	58
C	72

Student	Marks
D	81
E	67
F	90

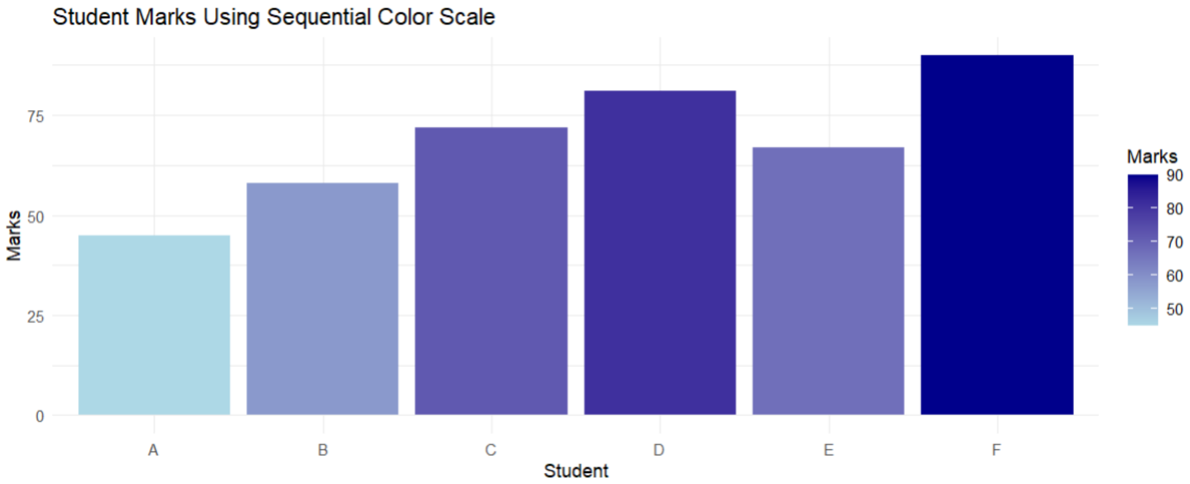
Questions

1. Write an R program to create a bar chart using a sequential color scale based on Marks.

R Code

```
library(ggplot2)

# Create dataset

data <- data.frame(

 Student = c("A","B","C","D","E","F"),

 Marks = c(45,58,72,81,67,90)

)

# Bar chart with sequential color scale

ggplot(data, aes(x = Student, y = Marks, fill = Marks)) +

 geom_bar(stat = "identity") +

 scale_fill_gradient(low = "lightblue", high = "darkblue") +

 labs(title = "Student Marks Using Sequential Color Scale",

     x = "Student",

     y = "Marks") +

 theme_minimal()
```

Result



A **bar chart** is generated where:

- The **X-axis** represents the students.
- The **Y-axis** represents the marks.
- The **bar colors** gradually change from light blue to dark blue according to the marks.
- Students with lower marks have lighter colors, while students with higher marks have darker colors.

---

2. Identify the students with the highest and lowest marks from the visualization.

Answer

From the bar chart, it is easy to identify the students with the highest and lowest marks based on the height and color intensity of the bars.

- **Student F** scored the **highest marks (90)**.
  - It has the tallest bar and the darkest blue color.
- **Student A** scored the **lowest marks (45)**.
  - It has the shortest bar and the lightest blue color.

The gradual change in color makes it easier to distinguish between higher and lower values without reading each numerical value.

---

3. Explain why a sequential color scale is appropriate.

Answer

A **sequential color scale** is appropriate because the dataset contains **continuous numerical values (Marks)**.

Reasons:

- Marks are measured on a continuous numerical scale.
- Sequential colors represent increasing values using gradually changing shades.
- Lighter colors indicate lower values.
- Darker colors indicate higher values.
- It helps users quickly identify patterns and trends.
- It improves readability and makes comparison easier.
- It avoids confusion by maintaining a logical progression of colors.

In this visualization:

- Student A (45) appears in the lightest shade.
- Student F (90) appears in the darkest shade.
- Intermediate values are represented using medium shades.

Therefore, a sequential color scale accurately represents the variation in marks.

---

4. Suggest a color palette suitable for color-blind users.

Answer

A suitable color palette for color-blind users is the **Viridis** color palette.

Reasons

- It is designed to be easily distinguishable by people with different types of color blindness.
- It provides a smooth transition from low to high values.
- It maintains good contrast.
- It is suitable for both digital screens and printed documents.
- It is widely used in scientific and statistical visualizations.

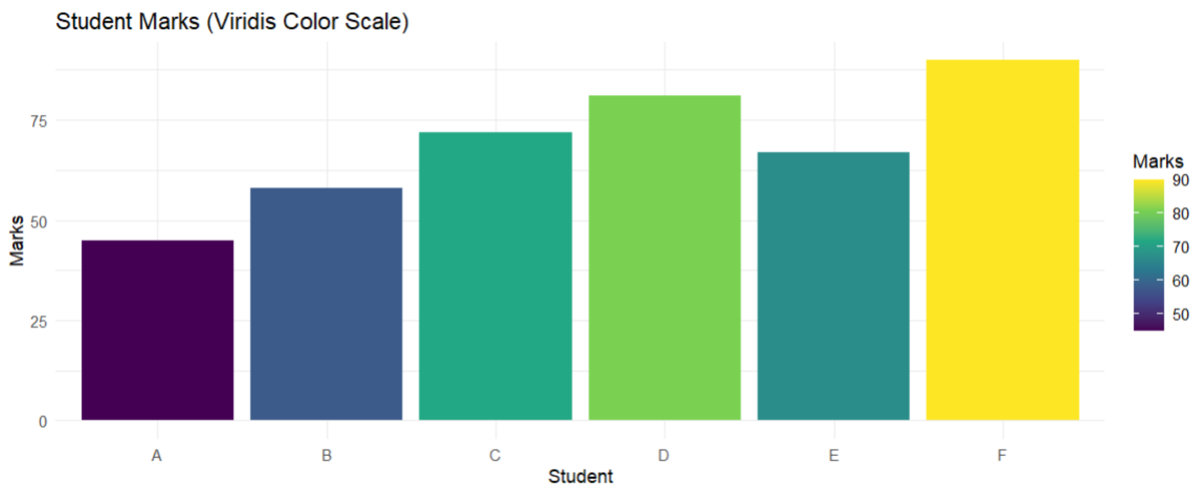
R Code using Viridis

library(ggplot2)

library(viridis)

```r
data <- data.frame(
  Student = c("A","B","C","D","E","F"),
  Marks = c(45,58,72,81,67,90)
)
ggplot(data, aes(Student, Marks, fill = Marks)) +
  geom_bar(stat = "identity") +
  scale_fill_viridis_c() +
  labs(title = "Student Marks (Viridis Color Scale)") +
  theme_minimal()
```

The **Viridis** palette is recommended because it ensures accessibility and enhances visualization quality.



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5. Explain the difference between sequential and categorical color scales.

Answer

Both sequential and categorical color scales are used for different types of data.

Sequential Color Scale

- Used for **continuous numerical data**.
- Colors change gradually from light to dark.
- Represents increasing or decreasing values.
- Suitable for marks, sales, profit, rainfall, temperature, etc.

- Helps identify trends and magnitude.

Example:

Marks	Color
45	Light Blue
58	Medium Blue
72	Blue
81	Dark Blue
90	Very Dark Blue

Categorical Color Scale

- Used for **categorical (discrete) data**.
- Each category is assigned a completely different color.
- No relationship exists between the colors.
- Used for variables like Department, Region, Gender, Product Category, etc.

Example:

Category	Color
Electronics	Red
Clothing	Blue
Furniture	Green

Comparison Table

Sequential Color Scale	Categorical Color Scale
Used for continuous numerical data	Used for categorical data
Shows gradual color changes	Uses distinct colors
Represents magnitude	Represents categories

Sequential Color Scale	Categorical Color Scale
Indicates increasing/decreasing values	Differentiates groups
Example: Marks, Sales	Example: Region, Category

Conclusion

Since **Marks** are continuous numerical values, a **sequential color scale** is the most appropriate choice. A **categorical color scale** would not effectively show the progression of marks because it is designed to distinguish categories rather than represent numerical magnitude.

Question 4: Highlighting with Color

Dataset

Region	Revenue (₹ Crores)
North	820
South	760
East	790
West	1010
Central	770

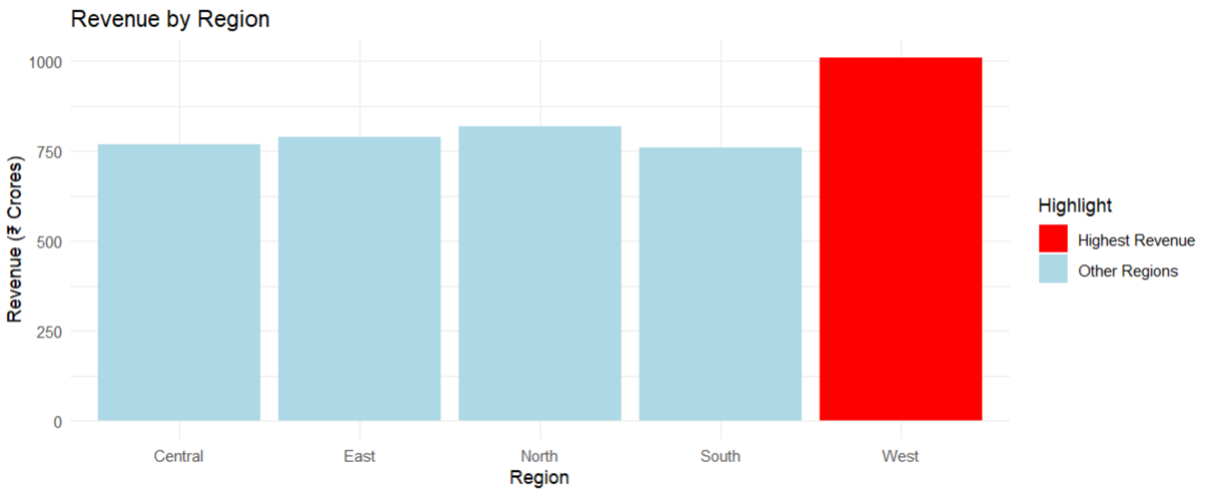
1. Write an R program to create a bar chart highlighting the region with the highest revenue.

R Code

```
library(ggplot2)

# Create dataset

data <- data.frame(

  Region = c("North","South","East","West","Central"),

  Revenue = c(820,760,790,1010,770)

)
```

```r
# Highlight the highest revenue region
data$Highlight <- ifelse(data$Revenue == max(data$Revenue),
                         "Highest Revenue",
                         "Other Regions")
# Create bar chart
ggplot(data, aes(x = Region, y = Revenue, fill = Highlight)) +
  geom_bar(stat = "identity") +
  scale_fill_manual(values = c("Highest Revenue" = "red",
                               "Other Regions" = "lightblue")) +
  labs(title = "Revenue by Region",
       x = "Region",
       y = "Revenue (₹ Crores)") +
  theme_minimal()
```

Result



A **bar chart** is displayed showing the revenue of all five regions.

- The **West** region is highlighted in **red** because it has the highest revenue.
- The remaining regions are shown in **light blue**.
- The visualization makes it easy to identify the highest-performing region at a glance.

---

2. Explain how highlighting improves data interpretation.

Answer

Highlighting is a visualization technique used to draw the viewer's attention to the most important information in a graph. By using a different color, size, or style, important data points can be identified immediately without carefully examining every value.

In this visualization, the **West** region is highlighted in red, making it stand out from the other regions.

Benefits of Highlighting

- It immediately attracts the viewer's attention to important data.
- It makes comparisons easier by emphasizing key values.
- It improves readability by reducing the time needed to interpret the chart.
- It helps users identify trends, patterns, or exceptional values quickly.
- It supports better decision-making by clearly displaying significant information.

Without highlighting, users would need to compare every bar individually to determine the highest revenue. Highlighting simplifies this process and enhances the overall effectiveness of the visualization.

---

3. Identify the highest-performing region.

Answer

From the given dataset:

Region	Revenue (₹ Crores)
North	820
South	760
East	790
West	**1010**
Central	770

The **West** region has the **highest revenue**, earning **₹1010 Crores**.

It is considered the highest-performing region because its revenue is greater than that of all the other regions.

The bar chart clearly shows the West region with the tallest bar, which is highlighted in red, making it easy to identify.

---

4. Discuss how excessive use of colors can affect interpretation.

Answer

Although colors improve the appearance of a visualization, using too many colors can reduce its effectiveness and make the graph difficult to understand.

Effects of Excessive Colors

- It creates confusion because viewers cannot easily distinguish which colors are important.
- Too many colors distract attention from the key information.
- It reduces the visual impact of highlighted data.
- Similar colors may be difficult to differentiate.
- It makes the chart look cluttered and less professional.
- People with color vision deficiencies may struggle to interpret the visualization.
- It can increase cognitive load, requiring more effort to understand the graph.

Best Practice

Only important information should be highlighted using a contrasting color, while the remaining data should use neutral or consistent colors. This ensures the chart remains simple, clear, and easy to interpret.

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5. Suggest one alternative highlighting technique other than color.

Answer

Apart from using color, there are several effective techniques to highlight important information in a chart.

One suitable alternative is **adding data labels**.

Explanation

Data labels display the exact values directly on the bars, allowing viewers to identify important information without relying on colors.

For example, the revenue value **1010** can be displayed above the West region's bar, making it immediately recognizable as the highest value.

Other Alternative Highlighting Techniques

- Increase the size or width of the highlighted bar.
- Use bold outlines around the important bar.
- Add annotations or text labels.
- Use arrows pointing to the important value.
- Sort the bars in descending order.
- Add a reference line indicating the average revenue.

These techniques improve interpretation while ensuring the chart remains accessible to users who may have difficulty distinguishing colors.

Question 5: Integrated Visualization

Dataset

City	Population (Million)	AQI	Rainfall (mm)	Region
A	8.2	110	850	North
B	5.6	180	430	South
C	9.4	90	1200	North
D	6.8	240	520	East
E	4.9	150	680	West

Questions

1. Write an R program to create a bubble chart using appropriate aesthetic mappings.

R Code

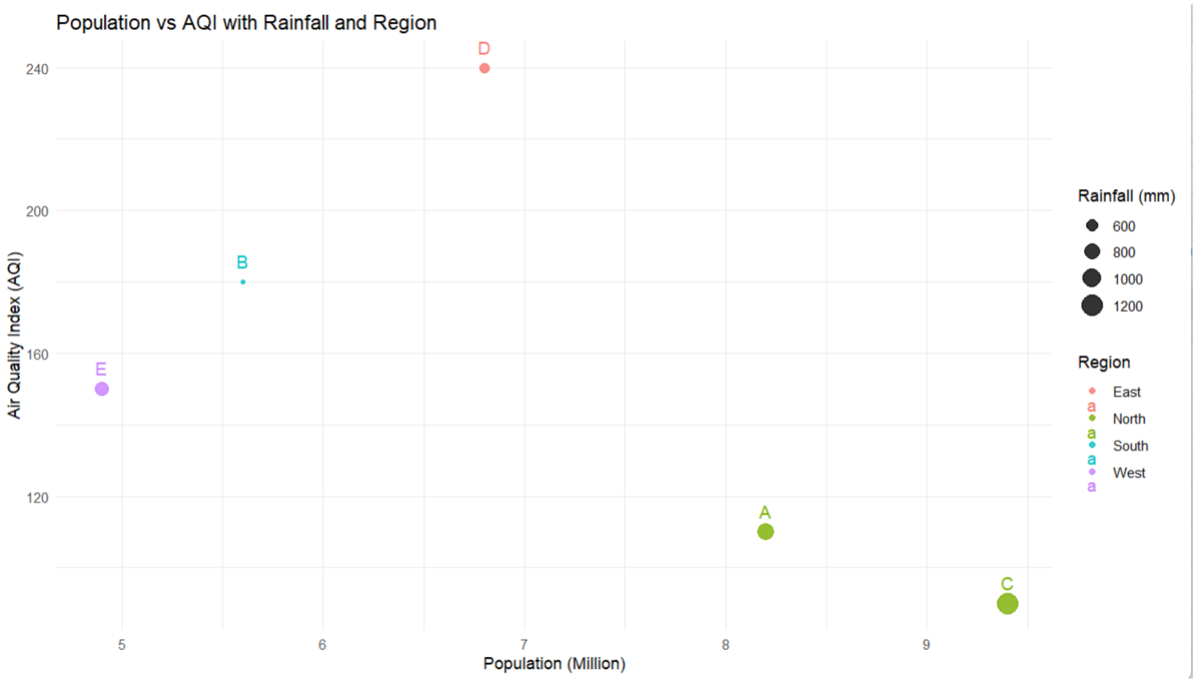
library(ggplot2)

Create dataset

data <- data.frame(

```
 City = c("A", "B", "C", "D", "E"),
 Population = c(8.2, 5.6, 9.4, 6.8, 4.9),
 AQI = c(110, 180, 90, 240, 150),
 Rainfall = c(850, 430, 1200, 520, 680),
 Region = c("North", "South", "North", "East", "West")
)
# Create Bubble Chart
ggplot(data,
    aes(x = Population,
        y = AQI,
        size = Rainfall,
        color = Region)) +
 geom_point(alpha = 0.8) +
 geom_text(aes(label = City),
        vjust = -1,
        size = 4) +
 labs(title = "Population vs AQI with Rainfall and Region",
    x = "Population (Million)",
    y = "Air Quality Index (AQI)",
    size = "Rainfall (mm)",
    color = "Region") +
 theme_minimal()
```

Result



The R program generates a **Bubble Chart** where:

- **Population** is displayed on the X-axis.
- **AQI (Air Quality Index)** is displayed on the Y-axis.
- **Rainfall** is represented by the size of each bubble.
- **Region** is represented by different bubble colors.
- **City names** are displayed above each bubble for better readability.

This visualization enables multiple variables to be analyzed simultaneously.

---

2. Select suitable aesthetics for Population, AQI, Rainfall, and Region.

Answer

Aesthetic mappings define how variables are represented visually in a graph. For this dataset, the following mappings are most appropriate:

Variable	**Aesthetic Used**	**Reason**
Population	X-axis	Population is a continuous numerical variable, making it suitable for the horizontal axis.
AQI	Y-axis	AQI is a continuous numerical variable and is best represented on the vertical axis.

Variable	Aesthetic Used	Reason
Rainfall	Bubble Size	Bubble size effectively represents the magnitude of rainfall. Larger bubbles indicate higher rainfall.
Region	Color	Different colors help distinguish cities belonging to different regions.
City	Text Labels	Labels identify each city without referring back to the dataset.

Using these aesthetic mappings allows four variables to be represented clearly within a single visualization, making the chart informative and easy to interpret.

---

3. Interpret the relationship between Population and AQI.

Answer

The bubble chart helps analyze the relationship between **Population** and **AQI (Air Quality Index)**.

From the dataset:

City	Population	AQI
A	8.2	110
B	5.6	180
C	9.4	90
D	6.8	240
E	4.9	150

Interpretation

- There is **no strong positive or negative relationship** between Population and AQI.
- City **C** has the highest population (9.4 million) but a relatively low AQI (90), indicating better air quality.
- City **D** has a moderate population (6.8 million) but the highest AQI (240), indicating poorer air quality.

- City **E** has the smallest population (4.9 million) but still has a moderately high AQI (150).

These observations indicate that population alone does not determine air quality. Other factors, such as industrial activities, traffic congestion, urban planning, weather conditions, and pollution control measures, also influence AQI.

Therefore, the visualization suggests that **Population and AQI do not exhibit a clear linear relationship** in this dataset.

---

4. Identify the continuous and categorical variables.

Answer

The dataset consists of both **continuous** and **categorical** variables.

Continuous Variables

Continuous variables are numerical values measured on a scale and can take many possible values.

- Population (Million)
- AQI (Air Quality Index)
- Rainfall (mm)

Categorical Variables

Categorical variables represent groups or labels rather than numerical measurements.

- City
- Region

Summary Table

Variable	Type
Population	Continuous
AQI	Continuous
Rainfall	Continuous
City	Categorical
Region	Categorical

Understanding the variable type helps in selecting appropriate visualization techniques and aesthetic mappings.

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5. Recommend one improvement to enhance the visualization's readability.

Answer

One effective improvement is to **add city labels** using geom_text() or geom_label().

Reasons

- Labels allow viewers to identify each city directly from the chart.
- They eliminate the need to refer back to the dataset.
- They make the visualization more informative and easier to interpret.
- They improve communication of important insights.

Other Possible Improvements

- Increase font size for better readability.
- Use the **Viridis** color palette to improve accessibility for color-blind users.
- Add grid lines to make value comparisons easier.
- Adjust bubble transparency (alpha) to reduce overlap.
- Include a descriptive legend for bubble size and color.
- Sort data if appropriate to improve comparison.
- Add annotations to highlight significant observations.

By implementing these improvements, the bubble chart becomes more visually appealing, accessible, and effective for data interpretation.
